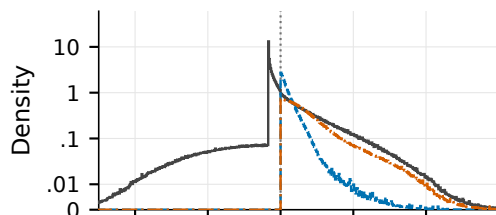


Exact-zero mass and nonzero activation distributions

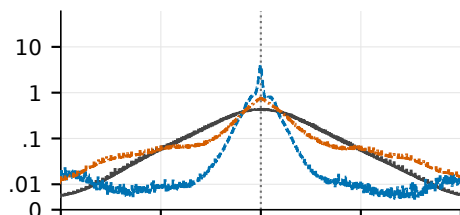
$\kappa=0$: FFN (h, m)

Exact zeros: A0 <0.01%
A4-OL1 67.22%; A7-OL1 64.52%



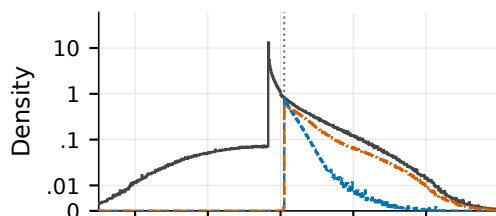
$\kappa=0$: Attention (q, k, v)

Exact zeros: A0 <0.01%
A4-OL1 <0.01%; A7-OL1 <0.01%



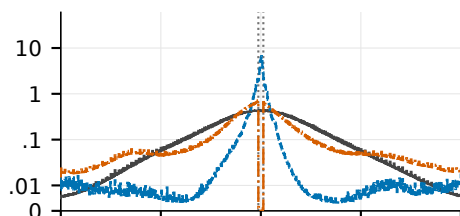
$\kappa=0.05$: FFN (h, m)

Exact zeros: A0 <0.01%
A4-OL1 86.71%; A7-OL1 75.29%



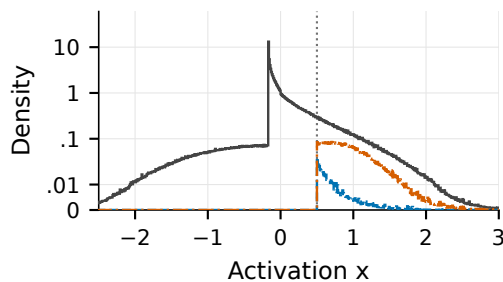
$\kappa=0.05$: Attention (q, k, v)

Exact zeros: A0 <0.01%
A4-OL1 <0.01%; A7-OL1 7.26%



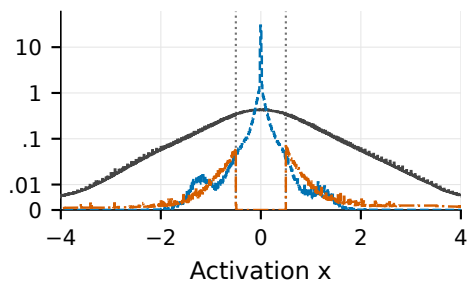
$\kappa=0.5$: FFN (h, m)

Exact zeros: A0 <0.01%
A4-OL1 99.31%; A7-OL1 93.64%



$\kappa=0.5$: Attention (q, k, v)

Exact zeros: A0 <0.01%
A4-OL1 0.22%; A7-OL1 95.60%



— A0 - - - A4-OL1 - . - A7-OL1 . . . Threshold